

ElastoSLM

Nested Low-Rank Elasticity for Small Language Models via Multi-Objective Budget Optimization

Supervisor: Sudaksh Kalra, University of Amsterdam · 6–9 months · Builds on Elastoformer (SEC '25), FlexRank (ICML '26), MOSP (NeurIPS '25)

THE PROBLEM

Small language models (0.5–3B) are shipped as fixed-cost monoliths: one checkpoint, one compute footprint, regardless of how much battery, memory or latency headroom the device has, and regardless of whether the query is trivial or hard. Elastic models fix this by extracting many nested submodels from one shared weight set — but how they are trained hides a real gap.

THE GAP

State-of-the-art elastic training minimizes a weighted sum of per-budget losses, $\min_{\theta} \sum_k \alpha_k \ell_k(\theta)$. This is a **linear scalarization** of a multi-objective problem, and it provably cannot reach Pareto points in non-convex regions of the front — for *any* α . That matters precisely at aggressive budgets, where elastic models degrade sharpest and where edge deployment actually lives. Chen et al. (NeurIPS 2025) showed the analogous point for LLM pruning: naive multi-objective extensions of a strong single-objective pruner collapse from 7.68 to 8.72 and 142.51 average perplexity. Treating multiple objectives properly is not cosmetic.

WHAT THE STUDENT WILL BUILD

- **Stage A — Elastic SLM.** Activation-aware low-rank factorization of per-layer projections, with a capped maximum rank so storage stays below the dense base model at every mode. Rank truncation preserves head_dim, so RoPE and GQA survive intact — the decisive advantage over head-dimension pruning for language models.
- **Stage B — MOO reformulation (core).** Replace the weighted sum with an explicit vector objective over budgets. Two tracks compared head-to-head: common-descent methods (MGDA / CAGrad / Nash-MTL) and preference-conditioned hypernetworks that learn the whole front rather than K points.
- **Stage C — Two-axis nesting.** Port MOSP's core-support idea into rank space: a core rank block shared across capabilities plus capability-specific extensions, giving a model indexed jointly by budget β and preference λ . No existing method provides this.

RESEARCH QUESTIONS

#	Question	Settled by
RQ1	Does scalarization leave Pareto-optimal budget configurations unreachable at SLM scale, and does MOO recover them?	Hypervolume vs α -sweep at matched compute
RQ2	Which MOO family suits objectives that are compute budgets over shared nested weights?	ϵ -indicator, spread, training overhead
RQ3	Can budget nesting and capability nesting coexist in one weight set?	(β, λ) model vs single- λ at matched β
RQ4	Do parameter reductions become wall-clock gains for batch-1 decode on edge?	TTFT and TPOT reported separately, Jetson Orin

RQ1 is the pivot — and a negative answer is still publishable: it would tell the elastic-models community that the simple objective suffices, and the thesis re-centres on RQ3 and RQ4.

TIMELINE

WP	Months	Content and gate
WP0	M1	Reproduce the published elastic low-rank Pareto curve on GPT-2 and Llama-3.2-1B
WP1	M2–3	Stage A on two SLMs — gate: beats decomposition-plus-search without consolidation
WP2	M4–6	Stage B, the core contribution — gate: hypervolume gain, or a documented negative for RQ1
WP3	M6–8	Stage C two-axis nesting — truncatable to a single capability if time runs short
WP4	M8–9	Jetson Orin measurement, switch latency, KV-cache reuse; thesis and paper draft

WHO THIS SUITS

A student comfortable reading a method paper and reimplementing it, rather than one looking for a well-specified engineering task. Strong PyTorch; solid linear algebra (SVD and subspace reasoning are load-bearing, not incidental); some optimization background — prior exposure to multi-objective or multi-task optimization helps but can be acquired during WP2. Jetson or inference-serving experience is a bonus.

Resources: 8×RTX 3090 for training, Jetson Orin for deployment measurement; FineWeb-Edu, GSM8K and MBPP-style corpora; lm-eval-harness. Existing Elastoformer and FlexRank codebases available as a starting point.